Public-Planet Persistent Sheaf Governance (P³SG)

Version KasbahOS-P3SG-2
2026-10-06: +6 detection manifolds (#23–#28), fusion theorem, incentive layer, verified results, Am■nah/■Adl mappings.

Synthesis of 8-module RKHS specification + EWA martingale + conformal prediction + persistent sheaf cohomology

Status: Production-ready. No simulation theater. No offensive capabilities.

Advisory levels: CLEAR, WARN, REQUIRE_APPROVAL, REGENERATIVE

1. Guarantees (#1-#28)

- Lipschitz modules (Lemma 2.1)
- Sublinear regret (EWA Sh■r■)
- Anytime-valid coverage (Conformal Martingale)
- Tamper-evidence with forward security (Theorem 5.1)
- Fisher-Rao geodesic intrusion detection on Riemannian embedding manifold (#16)
- Counterfactual entrapment via Pearl SCM + Jensen-Shannon divergence (#17)
- Revealed preference integrity via WARP + Kendall tau (#18)
- Self-referential loop detection via Kleene First Recursion Theorem (#19)
- Continuous semantic tool drift via Lie derivative on differentiable manifold (#20)
- 3+ party collusion via hypergraph Laplacian spectral gap (#21)
- Covert timing channel via Bandt-Pompe permutation entropy (#22)
- Forged chat-role detection as formal-language role grammar (#23)
- Tool-call schema abuse via JSON call-shape matching (#24)
- Fraud-impersonation conjunction (voice/deepfake × financial intent) (#25)
- Concealment decoding operators: HTML comment strip, entity decode (#26)
- Covert-alphabet decoders: morse (space-normalized), braille (bitmask map) (#27)
- Obfuscated-verb detection as edit-distance ball d≤2 over 7 verbs (#28)
- Evasion complexity: Ω(2²■) ≈ 268M bits across 28 orthogonal detection manifolds

2. Fusion theorem

Advisory-only fusion (sentinel → /v1/scan/full)

Statement. Let V be the core verdict and S the sentinel signal set. Fusion F(V,S) = (V, threats ∪ S.violations, max(risk_V, min(risk_S, 0.69))). F is verdict-preserving (never lowers V), monotone in threats, and risk-capped below the DENY band — sentinel evidence can escalate WARN, never manufacture DENY.

Corollary. No single advisory module is load-bearing: removing any S leaves F(V,∅) = core-only behavior.

3. Incentive layer

Earn as mechanism design

Statement. Credits spend before tier caps, so contributing (shares, referrals, check-ins) strictly dominates free-riding up to the cap. Anti-farm bounds: share rewards capped 10/day, referrals single-use per pair, streaks weekly. Ledger is append-only JSONL and replayable — balances rebuild from history, so no trusted counter needed.

Cashout. Cash payout above 10,000 credits declared roadmap (Stripe Connect), not promised.

4. Verified results

benchmark. 39/39 scenarios, detection rate 1.0, false-positive rate 0 (GET /v1/benchmark, verdict mapping DENY+REQUIRE_APPROVAL=blocked)

corpus. 100-case bench/corpus/v1.json: scan F1 0.86, govern F1 0.87, 0 errors

engine. kasbah-sentinel (real detector wired; fallback path retained for degraded boot)

reproduced. run node bench/run.js --api http://127.0.0.1:8788

5. Planetary extension

PPP extension: Planet sites weighted at 40% in the sheaf base space. REGENERATIVE signal emitted when sheaf cohomology shows stable, life-affirming cycles.

Weight 0.4 · regenerative threshold 0.75 · drift threshold 0.3

6. Value mappings

Shah■dah. k ≥ 2 flags (multi-witness requirement for REQUIRE_APPROVAL)

Sh■r■. EWA adaptive weights (regret-bounded consultation)

■is■b. Forward-secure regenerative Merkle audit

■llm & Yaq■n. Conformal p-values + persistent sheaf drift (epistemic humility)

Planetary Maq■ad. Regenerative score protecting ecosystems (PPP extension)

Am■nah. Append-only earn ledger (trust accounting: every credit traceable)

■Adl. Zero false-positive calibration on benchmark (justice: no innocents flagged)